

Lecture Note

(Cross-sectional) Regression Analysis I

- (A) What You Should Know by Now
- (B) Linear Regression Model
- (C) Empirical Examples (Estimation, Interpretation, Testing)
- (D) Measurement Errors
- (E) Data Structures

(A) What You *Should* Know by Now

- (a) Three data structures
 - (b) Simple linear regression models, Multiple linear regression models
 - (c) Classical Linear Regression Model (CLRM), Assumptions of CLRM, Gauss Markov Assumptions
 - (d) Small (or finite) sample properties of OLS estimators, large sample properties of OLS estimators
 - (e) Calculating estimates and variances, Interpreting OLS results, Understanding hypothesis testing
 - (f) Consequences of violations of Assumptions, Remedies for Assumption violations
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- You should spend considerable amount of time on reviewing Chapters 2-6, and Appendices A-C from the textbook. We will use Stata in this course.

(B) Linear Regression Model

Linear Regression Model: $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u$,

where y is the dependent (or explained) variable, $x_1, \dots, x_k$ are independent (or explanatory) variables (or regressors) and u is the error term (or disturbance). The parameter β_0 is the intercept term and β_j ($j = 1, 2, \dots, k$) is the slope parameter.

The simplest multiple regression linear regression model is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u,$$

where β_1 measures the change in y on average with respect to x_1 , holding other factor fixed, and β_2 measures the change in y on average with respect to x_2 , holding other factors fixed.

Objectives:

- (a) Estimating the unknown parameters $\beta_0, \beta_1, \dots, \beta_k$, and their variances; understanding the statistical properties of the OLS estimators $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ and $\hat{\sigma}^2$.
- (b) Carrying out hypothesis testing; performing the t test, F test, LM test, χ^2 test, etc.

Question: How do we use the OLS method to estimate the unknowns?

Let $\{(x_{i1}, \dots, x_{ik}, y_i): i = 1, \dots, n\}$ denote a random sample of size n from the population:

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + u_i, \quad i = 1, 2, \dots, n$$

$\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$, the Ordinary Least Squares (OLS) estimates of β_0, β_1 , are defined to minimize the sum of squared residuals (SSR),

$$\min \sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_K x_{ik})^2$$

The first-order conditions of this optimization problem:

$$\begin{aligned} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\ \sum_{i=1}^n x_{i1} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\ \sum_{i=1}^n x_{i2} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\ \vdots \\ \sum_{i=1}^n x_{ik} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0. \end{aligned}$$

There are $(k + 1)$ linear equations and we can solve for $(k + 1)$ unknowns $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$

$$\hat{\beta} = (X'X)^{-1}X'y$$

where $\hat{\beta}$ is a $(k + 1) \times 1$ matrix, X is an $n \times (k + 1)$ matrix with rank $(k + 1)$, and y is a $n \times 1$.

A simple regression model is the special case of the multiple regression models when $j = 1$, i.e., $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i$, $i = 1, 2, \dots, n$.

The OLS estimators are as follows.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \text{ and } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

If we have a sample in hand, then we can plug the information (i.e., data) into the estimators (i.e., the formula) to get estimates (i.e., numerical values)

For a particular sample, we can obtain the $\hat{\beta}_j$ estimate. But in practice there is almost impossible that the $\hat{\beta}_j$ estimate is equal to the population parameter β_j . If so, does it mean it is useless to do the regression analysis?

Statistical Properties of OLS Estimators

The $\hat{\beta}_j$ estimator is a random variable and it is the function of a random sample $\{(x_i, y_i): i = 1, \dots, n\}$. So, we can study its statistical properties. Note that here we use $\hat{\beta}_j$ as a representative for $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots$, and $\hat{\beta}_k$.

Under certain assumptions, the $\hat{\beta}_j$ estimates on average is equal to the population parameter β_j , i.e., $E(\hat{\beta}_j) = \beta_j$.

Question: What are the assumptions needed for having this unbiasedness property?

Assumption 1: Linear in parameters

Assumption 2: Random sampling

Assumption 3: No perfect multicollinearity

Assumption 4: Zero conditional mean

Assumption 5: Homoskedasticity

Assumption 6: Normality

Variance for $\hat{\beta}_j$

Under Assumptions 1-5, conditional on the values of explanatory variables

$$Var(\hat{\beta}_j | x_1, x_2, \dots, x_k) = \frac{\sigma^2}{\sum_{i=1}^n (x_{ji} - \bar{x}_j)^2 (1 - R_j^2)} = \frac{\sigma^2}{SST_j (1 - R_j^2)}.$$

The population parameter σ^2 remains to be estimated.

Remarks:

If the explanatory variables are highly but not perfectly correlated, then applied researchers would see the following symptoms:

- (a) Small changes in the data produce wide swings in the parameter estimates.
- (b) Coefficients may have very high standard errors and low significance levels even though they are jointly significant and the R-squared for the regression is quite high.

In practice, we can use the $\hat{\sigma}^2$ estimate to replace the unknown σ^2

$$\hat{\sigma}^2 = \frac{1}{n - k - 1} \sum_{i=1}^n \hat{u}_i^2,$$

where $\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}$

So, the estimated variance is $\widehat{Var}(\hat{\beta}_j) = \frac{\hat{\sigma}^2}{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2 (1 - R_j^2)}$

Theorem: (Gauss Markov Theorem)

Under Assumptions 1-5, the OLS estimators $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_k$ are the best linear unbiased estimator of $\beta_0, \beta_1, \dots, \beta_k$.

(C) Empirical Examples (Estimation, Interpretation, Testing)

Running an OLS regression model

Example: Return to Education (WAGE2.dta)

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + \beta_4 married + \beta_5 black + \beta_6 south + \beta_7 urban + u$$

$$\hat{\beta} = (X'X)^{-1}X'y, \quad \widehat{Var}(\hat{\beta}_j) = \frac{\frac{1}{n-k-1} \sum_{i=1}^n \hat{u}_i^2}{\sum_{i=1}^n (x_{ji} - \bar{x}_j)^2 (1 - R_j^2)}, \quad \text{where } j = 1, \dots, k$$

Source	SS	df	MS	Number of obs = 935		
Model	41.8377619	7	5.97682312	F(7, 927) = 44.75		
Residual	123.818521	927	.133569063	Prob > F = 0.0000		
Total	165.656283	934	.177362188	R-squared = 0.2526		
				Adj R-squared = 0.2469		
				Root MSE = .36547		

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.0654307	.0062504	10.47	0.000	.0531642	.0776973
exper	.014043	.0031852	4.41	0.000	.007792	.020294
tenure	.0117473	.002453	4.79	0.000	.0069333	.0165613
married	.1994171	.0390502	5.11	0.000	.1227801	.276054
black	-.1883499	.0376666	-5.00	0.000	-.2622717	-.1144281
south	-.0909036	.0262485	-3.46	0.001	-.142417	-.0393903
urban	.1839121	.0269583	6.82	0.000	.1310056	.2368185
_cons	5.395497	.113225	47.65	0.000	5.17329	5.617704

Standard and simple questions:

- How do we explain the R-squared value 0.2526 here?
- How do we interpret $\hat{\beta}_{educ} = 0.065$?
- How do we interpret the 95% confidence interval for β_{educ} ?
- Is the estimator $\hat{\beta}_{educ}$ an unbiased estimator? Why or why not?
- How do we fix or mitigate the problem if it is not an unbiased estimator?
- How do we reduce the standard errors?

Endogeneity Problem

Example: Return to Education (WAGE2.dta)

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + \beta_4 married + \beta_5 black + \beta_6 south + \beta_7 urban + \beta_8 IQ + u$$

Source	SS	df	MS	Number of obs = 935		
Model	43.5360162	8	5.44200202	F(8, 926) = 41.27		
Residual	122.120267	926	.131879338	Prob > F = 0.0000		
				R-squared = 0.2628		
				Adj R-squared = 0.2564		
Total	165.656283	934	.177362188	Root MSE = .36315		

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.0544106	.0069285	7.85	0.000	.0408133	.068008
exper	.0141458	.0031651	4.47	0.000	.0079342	.0203575
tenure	.0113951	.0024394	4.67	0.000	.0066077	.0161825
married	.1997644	.0388025	5.15	0.000	.1236134	.2759154
black	-.1431253	.0394925	-3.62	0.000	-.2206304	-.0656202
south	-.0801695	.0262529	-3.05	0.002	-.1316916	-.0286473
urban	.1819463	.0267929	6.79	0.000	.1293645	.2345281
IQ	.0035591	.0009918	3.59	0.000	.0016127	.0055056
_cons	5.176439	.1280006	40.44	0.000	4.925234	5.427644

We use *IQ* as the proxy variable for *ability*. What if we don't have suitable proxy variable for omitted variable *ability*?

Answer: Besides using (a) the proxy variable approach, we can (b) use Instrumental Variables (IV) approach to “substitute” *educ*, or (c) use Panel data model (e.g., fixed effect model) to eliminate *ability*.

Each approach requires **additional information**.

- (a) **Find** a proxy variable for *ability*
- (b) **Find** an IV for *educ*
- (c) **Re-interview** same group of people more than one period

Questions:

How do we know the homoskedasticity assumption is satisfied? Why is it so important to answer this question?

Answer: We can use the Breush-pagan test to test the homoskedasticity assumption.

H_0 : Homoskedasticity assumption holds H_1 : H_0 is wrong

Step1: *reg* $y \ x_1 \dots x_k$ (the main regression)

Step2: *predict* uhat , *residual* (to obtain the estimated residuals)

Step 3: generate $\text{uhat2} = \text{uhat} * \text{uhat}$ (create the uhat2 term as an dependent variable in the next regression)

Step 4: *reg* $\text{uhat2} \ x_1 \dots x_k$ (auxiliary regression and record the R-squared, $R_{\hat{u}^2}^2$)

Under the null, LM statistic = $n \times R_{\hat{u}^2}^2 \sim \chi_k^2$ (Breusch-Pagan test)

➤ If $LM > \chi_k^2$, we reject the null. Therefore, heteroskedasticity is present.

Recall the original regression model:

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + \beta_4 married + \beta_5 black + \beta_6 south + \beta_7 urban + \beta_8 IQ + u$$

The regression model used in the step 4:

$$\hat{u}^2 = \gamma_0 + \gamma_1 educ + \gamma_2 exper + \gamma_3 tenure + \gamma_4 married + \gamma_5 black + \gamma_6 south + \gamma_7 urban + \gamma_8 IQ + v$$

```
. reg uhat2 educ exper tenure married black south urban IQ
```

Source	SS	df	MS	Number of obs = 935		
Model	1.44972973	8	.181216216	F(8, 926) = 3.07		
Residual	54.7117297	926	.059083941	Prob > F = 0.0020		
Total	56.1614595	934	.060130042	R-squared = 0.0258		
				Adj R-squared = 0.0174		
				Root MSE = .24307		

uhat2	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	-.0018405	.0046375	-0.40	0.692	-.0109417	.0072608
exper	-.0009025	.0021185	-0.43	0.670	-.0050602	.0032552
tenure	-.006821	.0016328	-4.18	0.000	-.0100254	-.0036167
married	.0065477	.025972	0.25	0.801	-.0444232	.0575186
black	-.0219303	.0264339	-0.83	0.407	-.0738075	.0299469
south	.0312713	.0175721	1.78	0.075	-.0032145	.0657571
urban	.0081137	.0179335	0.45	0.651	-.0270814	.0433087
IQ	.0005953	.0006639	0.90	0.370	-.0007076	.0018981
_cons	.1353629	.0856758	1.58	0.114	-.0327784	.3035042

(a) LM statistic $= n \times R_{\hat{u}^2}^2 = 935 \times 0.0258 \approx 24.12 > \chi^2(8) = 2.773$. We reject the null at 5% significance level because the critical value is 2.773.

(b) We can use the F test to test for overall significance of the regression model.

$$F = \frac{R^2 / k}{\frac{1 - R^2}{n - k - 1}}$$

The F statistic value here is 3.07 and the corresponding p-value is 0.002. So we reject the null that all coefficients of the parameters are zeros, i.e., some explanatory variables can be used to explain the residual squared. In other words, heteroskedasticity is present.

Using robust standard estimation procedure, we can obtain the estimated robust standard errors

```
. reg lwage educ exper tenure married black south urban IQ, r
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Linear regression

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Number of obs =      935
F(   8,   926) =    48.51
Prob > F       =    0.0000
R-squared      =    0.2628
Root MSE      =    .36315
```

lwage	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.0544106	.007273	7.48	0.000	.0401372	.068684
exper	.0141458	.0032382	4.37	0.000	.0077908	.0205009
tenure	.0113951	.0025368	4.49	0.000	.0064166	.0163736
married	.1997644	.0390896	5.11	0.000	.1230499	.2764788
black	-.1431253	.0376459	-3.80	0.000	-.2170064	-.0692442
south	-.0801695	.0277381	-2.89	0.004	-.1346063	-.0257327
urban	.1819463	.0267419	6.80	0.000	.1294646	.234428
IQ	.0035591	.0009564	3.72	0.000	.0016822	.0054361
_cons	5.176439	.1212236	42.70	0.000	4.938534	5.414344

Remarks:

- (a) Even if we reject the homoskedasticity assumption, we usually do not know the structure of heteroskedasticity.
- (b) We could use the robust standard errors (especially when we have a fairly large sample size). Even if there is no heteroskedasticity, so long as the sample size is large enough we can still use the robust standard errors as they are appropriate even under homoskedasticity.
- (c) Usually, the robust standard errors are bigger than the uncorrected wrong standard errors.

Three types of null hypotheses

(1) Single Population parameter: the t test $H_0: \beta_j = a_j$

(2) Testing Hypotheses about a Single Linear Combination of the Parameters

$$H_0: \beta_2 = \beta_3 \Leftrightarrow H_0: \beta_2 - \beta_3 = 0$$

(3) Testing Multiple Linear Restriction: the F Test

$$H_0: \beta_{K-q-1} = 0, \dots, \beta_K = 0$$

(D) Measurement Errors

When there are measurement errors in the dependent variable (MEY), the OLS estimators are still unbiased and consistent so long as the measurement errors are not correlated with the explanatory variables. In general, we find no relationships between measurement errors and x variables on average.

When there are measurement errors in the explanatory variables (MEX), the OLS estimators are generally biased (and inconsistent) because measurement errors are usually correlated with the explanatory variable(s).

Exercise:

Consider a simple regression model $y_i = \beta_0 + \beta_1 x_{i1}^* + u_i$, $n = 1, \dots, n$. Suppose x_2^* is unobserved (e.g., actual income) and x_2 is a measure (e.g., reported income) for x_2^* . So, the observed data x_2 is imperfectly measured version of x_2^* . Let's assume $x_2 = x_2^* + MEX$, where the $MEX \sim N(0, \sigma_{MEX}^2)$ and is uncorrelated with the

unobserved measure x_2^* . Show that the OLS estimator $\hat{\beta}_1$ is biased and inconsistent, that is $\hat{\beta}_1 \approx \beta_1 \left(\frac{\sigma_{x_2^*}^2}{\sigma_{MEX}^2 + \sigma_{x_2^*}^2} \right)$

when $n \rightarrow \infty$.

(E) Data Structures

Cross-sectional Data: a sample that consists of many individuals (or households or firms or cities or countries, etc) taken at a same point of time.

- (a) Random sampling from the underlying population
- (b) Each observation (aka cross-section unit) corresponds to a person or a firm, etc.
- (c) Change the order of observations won't change the information/estimation results. (Refer to point (a))

Time series data: consists of a sequence of observations which are ordered in time

- (a) Observing only one entity; it can be a country, a firm or a person, etc.
- (b) Track down that particular entity over a period of time, say 15 years. For example, Singapore quarterly GDP from 1980 Q1 -2015 Q4, i.e., 144 quarterly GDP observations and each observation corresponds to the quarter observed
- (c) A time series data has a natural temporal ordering. Observations are always dependent across time, or “serial correlated”

Panel data: consists of a time series for *each* cross-sectional unit (or entity)

- (a) Contains multiple time periods for the cross-sectional units